

Le système international d'unités The International System of Units

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Annexe 2

Annex 2

Valeurs recommandées des fréquences étalons
destinées à la mise en pratique de la définition du
mètre et aux représentations secondaires de la
seconde

**Recommended values of standard frequencies
for applications including the practical
realization of the metre and secondary
representations of the second**

Partie 2

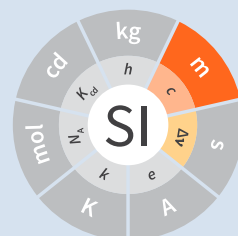
Part 2

Hydrogen Atom ()

Hydrogen Atom ()

CCL-CCTF Frequency Standards Working Group

May 2016
mai 2016



**Recommended values of standard
frequencies for applications including
the practical realization of the metre and
secondary representations of the second
Part 2: Hydrogen Atom ()**

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1. Recommended value [CIPM REC 2 \(2015, E\)](#) of the frequency

$$f(^1\text{H}) = 1\,233\,030\,706\,593\,514\text{ Hz}$$

equivalent to

$$\lambda(^1\text{H}) = 243\,134\,624.626\,044\text{ fm},$$

with an estimated relative standard uncertainty of 9×10^{-15} .

The 1S–2S transition is excited by two photons of the frequency 1233 THz

2. Source data

Adopted value $f(^1\text{H}) = 1\,233\,030\,706\,593\,514\text{ Hz}$
 $u_c/y = 9 \times 10^{-15}$

calculated from

$2 \times f(^1\text{H})/\text{Hz}$	u/Hz	source data
2 466 061 413 187 035	10	[2]
2 466 061 413 187 018	11	[3]

as a weighted mean of the above values divided by two.

Given the noticeable drift of the values of the past measurements [3] and the fact that the values come from a single laboratory the CCTF considered it prudent to attribute an increased standard uncertainty by a factor of three.

^1H neutral atom, $1\text{S} - 2\text{S}$ unperturbed optical transition

References

- [1] CIPM REC 2 (2015, E) International Committee for Weights and Measures (2015) *Updates to the list of standard frequencies* (Paris: International Bureau of Weights and Measures) https://www.bipm.org/en/committees/ci/cipm/104-_1-2015/resolution-2.
- [2] C. G. Parthey, A. Matveev, J. Alnis, B. Bernhardt, A. Beyer, R. Holzwarth, A. Maistrou, R. Pohl, K. Predehl, T. Udem, T. Wilken, N. Kolachevsky, M. Abgrall, D. Rovera, C. Salomon, P. Laurent, and T. W. Hänsch: Improved Measurement of the Hydrogen 1S—2S Transition Frequency, *Phys. Rev. Lett.* **107**, 203001 (2011).
- [3] A. Matveev, Ch. G. Parthey, K. Predehl, J. Alnis, A. Beyer, R. Holzwarth, Th. Udem, T. Wilken, N. Kolachevsky, M. Abgrall, D. Rovera, Ch. Salomon, Ph. Laurent, G. Grosche, O. Terra, Th. Legero, H. Schnatz, S. Weyers, B. Altschul, Th. W. Hänsch: Precision Measurement of the Hydrogen 1S—2S Frequency via a 920— km Fiber Link, *Phys. Rev. Lett.* **110**, 230801 (2013).